ic_autolab round 30 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T10:00:06Z at git `897e546fea307e6e6a2ebb4788912aea514b9c6f`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-2ed644` (`python3 tools/pcc/pcc.py standing PCC-20261104-2ed644`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
24	**no_challenger**	none (best candidate `wsum2r-b30-n2`, train S 3.063 / 3.385 / 6.283)	-	-	-	-	-	2.738 / 2.881 / 4.165	1.191 / 1.298 / 2.264	inc: 2.3x / 2.2x / 1.8x
28	**no_challenger**	none (best candidate `wsum2r-b60-n2`, train S 2.713 / 2.992 / 5.509)	-	-	-	-	-	2.277 / 2.377 / 3.279	0.945 / 1.034 / 1.827	inc: 2.4x / 2.3x / 1.8x
32	**no_challenger**	none (best candidate `wsum2r-b120-n2`, train S 6.017 / 6.967 / 15.515)	-	-	-	-	-	2.513 / 2.752 / 4.899	0.844 / 0.927 / 1.671	inc: 3.0x / 3.0x / 2.9x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

- k = 24: no_challenger
- k = 28: no_challenger
- k = 32: no_challenger

Train (development screen, in sample)

k = 24: session `ECBS1hbedcd44f7d06` (run exit 0, verify exit 0, receipt sha256 `caf928122bbc3ed1cccdfbffebecd93060a99a83c9dd0694587ddc3ec305d7a0`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	32	32	1.191 / 1.298 / 2.264	matched reference
`inc`	wsum2g-b30	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=30,free=1`	32	32	2.738 / 2.881 / 4.165	incumbent
`wsum2r-b30-n2`	wsum2r-b30	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=30,free=1`	32	32	3.063 / 3.385 / 6.283	exploration: family wsum2r-b30 untried at k = 24, N = 0.0001 x N_a
`wsum2r-b45-n2`	wsum2r-b45	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=45,free=1`	32	32	3.885 / 4.057 / 5.603	exploration: family wsum2r-b45 untried at k = 24, N = 0.0001 x N_a
`wsum2r-b60-n2`	wsum2r-b60	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=60,free=1`	32	32	5.786 / 5.899 / 6.914	exploration: family wsum2r-b60 untried at k = 24, N = 0.0001 x N_a
`wsum2r-b90-n2`	wsum2r-b90	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=90,free=1`	32	32	11.885 / 11.966 / 12.698	exploration: family wsum2r-b90 untried at k = 24, N = 0.0001 x N_a
`wsum2r-b120-n2`	wsum2r-b120	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=120,free=1`	32	32	20.510 / 20.568 / 21.091	exploration: family wsum2r-b120 untried at k = 24, N = 0.0001 x N_a

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
wsum2r-b160-n2	wsum2r-b160	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=160,free=1	32	32	35.947 / 35.988 / 36.355	exploration: family wsum2r-b160 untried at k = 24, N = 0.0001 x N_a

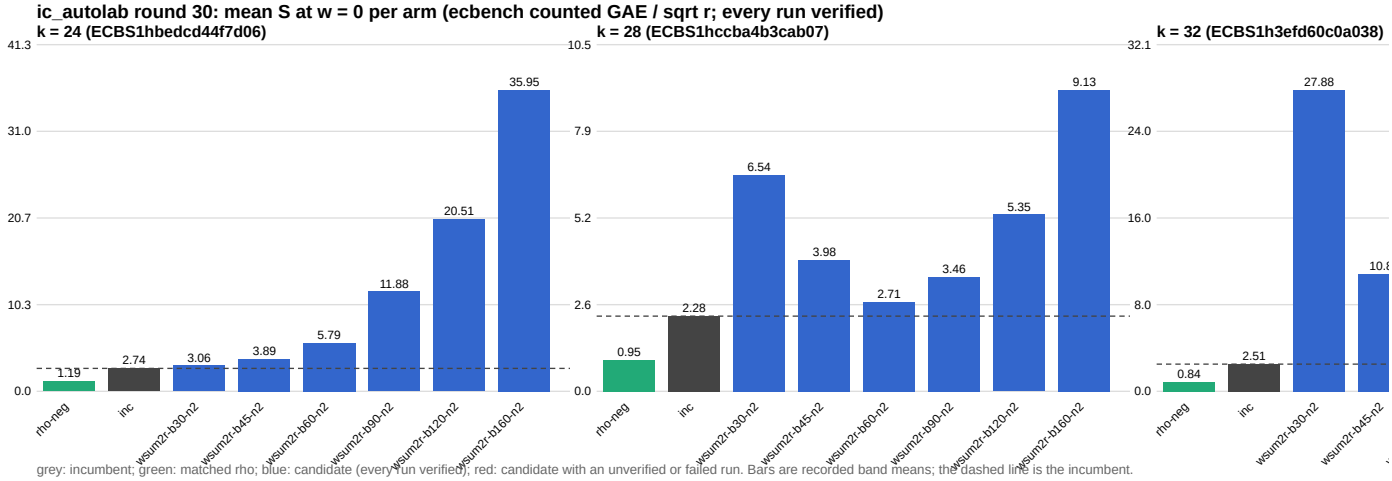
k = 28: session `ECBS1hccba4b3cab07` (run exit 0, verify exit 0, receipt sha256 `a46494be9ad8f47ddc5de47f43e1ded3151c4e9f34240270e0fe17683c5dc7ad`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	32	32	0.945 / 1.034 / 1.827	matched reference
inc	wsum2g-b60	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=60,free=1	32	32	2.277 / 2.377 / 3.279	incumbent
wsum2r-b30-n2	wsum2r-b30	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=30,free=1	32	32	6.544 / 7.778 / 18.883	exploration: family wsum2r-b30 untried at k = 28, N = 0.0001 x N_a
wsum2r-b45-n2	wsum2r-b45	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=45,free=1	32	32	3.981 / 4.624 / 10.411	exploration: family wsum2r-b45 untried at k = 28, N = 0.0001 x N_a
wsum2r-b60-n2	wsum2r-b60	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=60,free=1	32	32	2.713 / 2.992 / 5.509	exploration: family wsum2r-b60 untried at k = 28, N = 0.0001 x N_a
wsum2r-b90-n2	wsum2r-b90	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=90,free=1	32	32	3.456 / 3.572 / 4.612	exploration: family wsum2r-b90 untried at k = 28, N = 0.0001 x N_a
wsum2r-b120-n2	wsum2r-b120	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=120,free=1	32	32	5.354 / 5.412 / 5.935	exploration: family wsum2r-b120 untried at k = 28, N = 0.0001 x N_a
wsum2r-b160-n2	wsum2r-b160	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=160,free=1	32	32	9.128 / 9.165 / 9.495	exploration: family wsum2r-b160 untried at k = 28, N = 0.0001 x N_a

k = 32: session `ECBS1h3efd60c0a038` (run exit 0, verify exit 0, receipt sha256 `40078c39388e9ab9181c7e6dbae215cf72f83542cbfd39c876021fc53138398e`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	32	32	0.844 / 0.927 / 1.671	matched reference
inc	wsum2-b120	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=120,free=1	32	32	2.513 / 2.752 / 4.899	incumbent
wsum2r-b30-n2	wsum2r-b30	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=30,free=1	32	32	27.880 / 33.435 / 83.428	exploration: family wsum2r-b30 untried at k = 32, N = 0.0001 x N_a
wsum2r-b45-n2	wsum2r-b45	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=45,free=1	32	32	10.876 / 13.012 / 32.238	exploration: family wsum2r-b45 untried at k = 32, N = 0.0001 x N_a
wsum2r-b60-n2	wsum2r-b60	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=60,free=1	32	32	8.045 / 9.588 / 23.473	exploration: family wsum2r-b60 untried at k = 32, N = 0.0001 x N_a
wsum2r-b90-n2	wsum2r-b90	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=90,free=1	32	32	7.170 / 8.459 / 20.065	exploration: family wsum2r-b90 untried at k

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`wsum2r-b120-n2`	wsum2r-b120	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=120,free=1`	32	32	6.017 / 6.967 / 15.515	exploration: family wsum2r-b120 untried at k = 32, N = 0.0001 x N_a
`wsum2r-b160-n2`	wsum2r-b160	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=160,free=1`	32	32	6.270 / 7.076 / 14.331	exploration: family wsum2r-b160 untried at k = 32, N = 0.0001 x N_a



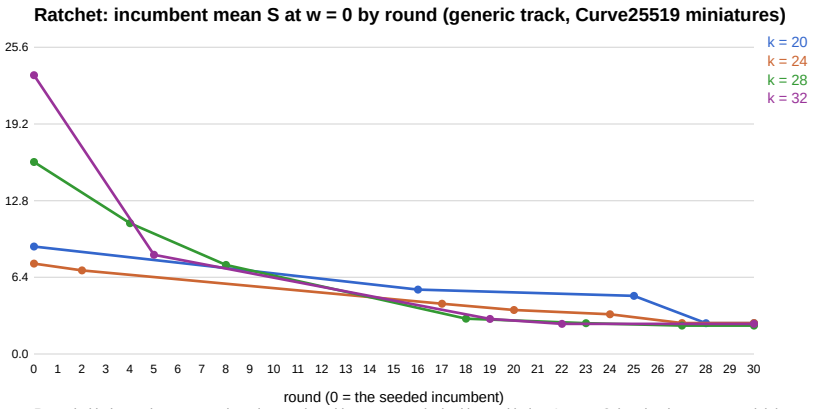
train screen

Selection

- k = 24: no_challenger (Pareto alternative at w = 1: `wsum2r-b45-n2`)
- k = 28: no_challenger (Pareto alternative at w = 1: `wsum2r-b90-n2`)
- k = 32: no_challenger (Pareto alternative at w = 1: `wsum2r-b160-n2`)

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`wsum2g-b15-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=15,free=1`)	28	2.575 / 2.691 / 3.741	1.8x / 1.7x / 1.5x
24	`wsum2g-b30-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=30,free=1`)	27	2.584 / 2.711 / 3.855	2.0x / 2.0x / 1.6x
28	`wsum2g-b60-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=60,free=1`)	27	2.373 / 2.483 / 3.472	2.4x / 2.3x / 1.8x
32	`wsum2-b120-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=120,free=1`)	22	2.519 / 2.759 / 4.915	2.7x / 2.7x / 2.7x



ratchet

Evidence and deviations

- ecbench binary sha256: `6229f85b81f9a2e8f2811d675f77f4f830ca536422a1e92680a382183450a62c`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.